CompositionMeasure for RenyiDivergence

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of CompositionMeasure for RenyiDivergence in mod.rs at commit f5bb719 (outdated[1]).

1 Hoare Triple

Precondition

Compiler-verified

Types matching pseudocode.

Caller-verified

None

Pseudocode

```python
class CompositionMeasure(RenyiDivergence):
    def composability(  #
        self, adaptivity: Adaptivity
    ) -> Composability:
        return Composability.Concurrent

    def compose(self, d_mids: Vec[Self_Distance]) -> Self_Distance:
        # curves sharing an allocation are grouped as one (curve, k) pair,
        # in first-occurrence order, so that each distinct curve is
        # evaluated once, not once per copy
        groups = []  # Vec<(Self_Distance, u32)>
        for d_mid in d_mids:
            group = next(
                (group for group in groups
                 if Arc.ptr_eq(group[0].function, d_mid.function)),
                None,
            )
            if group is not None:
                group[1] += 1
            else:
                groups.push((d_mid, 1))

        def curve(alpha: float) -> float: #
            epsilons = [
                d_mid(alpha).inf_mul(k)  #
                for d_mid, k in groups
```

[1]See new changes with `git diff f5bb719..9911cc6a rust/src/combinators/sequential_composition/mod.rs`

```
27              ]
28
29              d_out = 0.0
30              for d_mid in epsilons:
31                  d_out = d_out.inf_add(d_mid)
32              return d_out
33
34          return Function.new_fallible(curve)
```

Postcondition

Theorem 1.1. `composability` returns `Ok(out)` if the composition of a vector of privacy parameters `d_mids` is bounded above by `self.compose(d_mids)` under `adaptivity` adaptivity and `out`-composability. Otherwise returns an error.

Proof. Since curves that share an allocation are the same function, the grouping on line 11 only changes the representation of `d_mids`: a curve appearing `k` times is stored once with multiplicity `k`, in first-occurrence order. The new curve constructed on line 23 composes all epsilon parameters at a given fixed `alpha`, evaluating each distinct curve once. By the postcondition of `InfMul` on line 25, $\texttt{k} \cdot \texttt{d\_mid}(\texttt{alpha}) \leq \texttt{d\_mid}(\texttt{alpha}).\texttt{inf\_mul}(\texttt{k})$. Then by the postcondition of `InfAdd` we have that for any choice of `alpha`, $\sum_i \texttt{d\_mid}_i(\texttt{alpha}) \leq \texttt{compose(d\_mids)}(\alpha)$.

Adaptivity	Sequential	Concurrent
Non-Adaptive	Proposition 1[Mir17]	Theorem 2[Lyu22]
Adaptive	-	-
Fully-Adaptive	Theorem 4.3[FZ22]	Theorem 1.22[VW21]

This table is reflected in the implementation of `composability` on line 2.

□

References

[FZ22] Vitaly Feldman and Tijana Zrnic. Individual privacy accounting via a renyi filter, 2022.

[Lyu22] Xin Lyu. Composition theorems for interactive differential privacy, 2022.

[Mir17] Ilya Mironov. Rényi differential privacy. In *2017 IEEE 30th Computer Security Foundations Symposium (CSF)*, page 263–275. IEEE, August 2017.

[VW21] Salil Vadhan and Tianhao Wang. Concurrent composition of differential privacy, 2021.